

Ujaan Das

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EDUCATION

University of Warwick <i>Master of Science in Computer Science</i>	Coventry, United Kingdom <i>Sep. 2025 – Sep. 2026</i>
Hong Kong University of Science and Technology <i>Bachelor of Engineering in Computer Engineering</i>	Hong Kong SAR <i>Sep. 2021 – May 2025</i>
Northwestern University <i>Exchange Semester</i>	Illinois, USA <i>Jan. 2024 – Apr. 2024</i>
West Island School <i>International Baccalaureate (41/45; HL: Computer Science, Physics, Psychology)</i> <i>iGCSEs (English (A*), Mathematics (8), Sciences (9); Art (7), Business Studies (B), Computer Science (A), Spanish (A), Global Perspectives (A))</i>	Hong Kong SAR <i>Sep. 2019 – May 2021</i> <i>Sep. 2017 – May 2019</i>

EXPERIENCE

Software Engineering Intern (Part Time) <i>Stellerus Technology</i>	Jan. 2025 – Aug. 2025 <i>Hong Kong SAR</i>
- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session - Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30% - Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions	
Research Assistant <i>CASTLE Lab, HKUST</i>	May 2024 – Sep. 2024 <i>Hong Kong SAR</i>
- Engineered high-throughput semantic-analysis pipelines in Python and Java to extract semantic signals from 30k+ scraped GitHub repositories to inform research in LLM-assisted testcase generation - Integrated Tree-sitter to parse multi-million-line codebases into ASTs and CFGs, converting complex source code into lightweight, queryable intermediate representations - Optimized this parsing logic to cut per-project analysis time by 45%, ultimately increasing usable data yield by 3.2x to support large-scale experiments, slated for publication in ACM ICSE'27	
Software Engineering Intern <i>Stellerus Technology</i>	May 2023 – Aug. 2023 <i>Hong Kong SAR</i>
- Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times - Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34% - Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%	
Research Assistant <i>HCI Initiative, HKUST</i>	Jun. 2022 – Aug. 2022 <i>Hong Kong SAR</i>
- Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments - Collaborated with post-docs to build a modular React + TypeScript frontend using custom hooks and Tailwind CSS, improving component reusability and streamlining development across multiple researchers - Contributed to FARPLS, a feature-augmented preference learning system that improved label quality and user engagement without increasing cognitive load (published at ACM/IUI'24)	

PROJECTS

MSc Dissertation - Invariants <i>C++, Gtest, CMake, Python</i>	Github
- Applying formal systems theory to LLMs for improved structured output generation via runtime constraints - Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines	
Snip <i>C++, Termios, Gtest, CMake, Nix</i>	Github
Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture	
yumcha <i>Go, C, CGo, Cocoa, Nix</i>	Github

- Engineered a macOS tiling window manager in Go and C using Cocoa/CoreGraphics, implementing an i3-style tree layout engine for dynamic, scriptable window management without disabling SIP

Undergraduate FYP - Follow-Me Robot | *C++, Python, ROS2, Raspberry Pi, Docker*

[Github](#)

- Developed a real-time UWB tracking system via custom ROS2 Python/C++ nodes, applying multi-rate Kalman and EMA filtering for stable sub-centimeter localization under noisy conditions

oojsite | *Go, HTML, TailwindCSS, Nix*

[Github](#)

- Built a static site generator in Go to compile Markdown into composable HTML, actively used for [my blog](#)

TerraViz | *Next.js, React, TypeScript, CesiumJS, TailwindCSS*

[DevPost](#)

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data

SKILLS

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: GTest, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git